

Karnatak University Mathematics Teachers
Association (KUMTA)

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Mathematics Laboratory Manual

Paper – Differential Equations and Integral Transforms

Prepared by – Dr. Shashikant, JSS College, Dharwad

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PROGRAM 1: Program to verify the exactness of a differential equation.

Program:

a)

```
/*Program to verify the exactness of a differential equation */
```

```
kill(all)$
```

```
Z: (x*y + y^2)*dx + (x^2 + 2*x*y)*dy;
```

```
M : coeff(Z, dx);
```

```
N : coeff(Z, dy);
```

```
My : diff(M, y);
```

```
Nx : diff(N, x);
```

```
if My = Nx then
```

```
    print("The given differential equation is exact")
```

```
else
```

```
    print("The given differential equation is not exact")$
```

Output:

```
Z : (xy + y2)dx + (x2 + 2xy)dy
```

```
M : xy + y2
```

```
N : x2 + 2xy
```

```
My : x + 2y
```

```
Nx : x + 2y
```

The given differential equation is exact

b)

*/*Program to verify the exactness of a differential equation */*

kill(all)\$

Z: (x*y)*dx + (x + y)*dy;

M : coeff(Z, dx);

N : coeff(Z, dy);

My : diff(M, y);

Nx : diff(N, x);

if My = Nx then

 print("The given differential equation is exact")

else

 print("The given differential equation is not exact")\$

Output:

Z : xydx + (x + y)dy

M: xy

N : x + y

My : x

Nx: 1

The given differential equation is not exact

PROGRAM 2: Program to get the solutions of differential equations that are solvable for x , y , p .

Program:

a)

```
/*Program to get the solutions of differential equations that are solvable for x */
kill(all)$
x: y+p$
x(y):= y +p(y)$
d: diff(x(y),y)$
eq:1/p - d$
q: solve(eq,'diff(p(y),y,1))$
sol1: integrate(1,y) = -integrate(p/(p-1),p)+C$
print("The parametric solution of the given problem is x =", x, "and", sol1)$
```

Output:

The parametric solution of the given problem is $x = y+p$ and $y = -p - \log(p-1) + C$

b)

```
/*Program to get the solutions of differential equations that are solvable for y */
kill(all)$
x: sin(p) - p*cos(p)$
y(x):=sin(p(x)) - p(x)*cos(p(x))$
d: diff(y(x),x)$
eq:p - d$ // based on manually
q: solve(eq,'diff(p(x),x,1))$
sol1: integrate(1,x) = integrate(sin(p),p)+C$ // based on manually
print("The parametric solution of the given problem is y =", y, "and", sol1)$
```

Output:

The parametric solution of the given problem is $y = \sin(p) - p\cos(p)$ and $x = C - \cos(p)$

c)

*/*Program to get the solutions of differential equations that are solvable for p*/*

kill(all)\$

*a:x^2*p^2+x*y*p-6*y^2\$*

solve(a,p)\$

f:subst(p='diff(y,x,1),%)\$

f1:f[1]\$

ode2(f1,y,x)\$

sol1:(lhs(%) - rhs(%));

f2:f[2]\$

ode2(f2,y,x)\$

sol2:(lhs(%) - rhs(%));

*print("The solution of the differential equation is ", sol1*sol2 = 0)\$*

Output:

Sol1 : $y - \%C/x^3$

Sol2 : $y - \%Cx^2$

The solution of the differential equation is $(y - \%C/x^3)(y - \%Cx^2) = 0$

PROGRAM 3: Program to find the Complementary Function and Particular Integral of linear differential equations with constant coefficients.

Program:

a)

```
/*Program to find the Complementary Function and Particular Integral of linear
differential equations with constant coefficients*/
kill(all)$
eqn: 'diff(y,x,2) - 5*'diff(y,x,1) + 6*y = sin(3*x)$
print("The given linear equation is:", eqn)$
sol: ode2(eqn,y,x)$
print("The solution of the given LDE is ", sol)$
```

Output 1:

The given linear equation is: $(d^2/dx^2)y - 5 (d/dx)y + 6y = \sin(3x)$

The solution of the given LDE is $y = -\frac{\sin(3x) - 5 \cos(3x)}{78} + \%k1\%e^{3x} +$
 $\%k2\%e^{-2x}$

Output 2: (eqn = 'diff(y,x,2) + 4*'diff(y,x,1) + 4*y = cos(2*x)\$)

The given linear equation is: $(d^2/dx^2)y - 4 (d/dx)y + 4y = \cos(2x)$

The solution of the given LDE is $y = -\frac{\sin(2x)}{20} + \frac{\cos(2x)}{20} + (\%k1 +$
 $\%k2 x)\%e^{-2x}$

PROGRAM 4: Program to get solutions of linear partial differential equations of type 1 to type 4 and Lagrange's method.

Program for type- I:

```
/* Partial differential equation of type-I */
kill(all) $
eqn: (p*q = 1);
z: a * x + b * y + c $
h: subst([p=a, q=b], eqn) $
solve(h, a);
h1: subst(%,z) $
print("The required solution is:")$
print(h1, "=0") $
```

Output :

```
pq = 1
[ a = 1/b]
The required solution is:
by + x/b + c = 0
```

Program for type- II:

```
/* Partial differential equation of type-II */
kill(all) $
eqn : p*(1-q^2) = q*(1-z);
eq1: subst([p='diff(z,u), q=a*'diff(z,u)],eqn);
h1: solve(eq1,'diff(z,u)) $
ode2(h1[2], z,u)$
h2: subst(u=x+a*y, %) $
print(" The required solution is:",h2) $
```

Output :

```
p(1-q^2) = q(1-z)
(dz/du)(1-a^2(dz/du)^2) = a(1-z)(dz/du)
The required solution is:   $2\sqrt{az - a + 1} = ay + x + \%c$ 
```

Program for type- III:

```
/* Partial differential equation of type-III */
kill(all) $
eqn : p + x = q + y;
rhs : p + x = k $
lhs : q + y = k $
P : solve(rhs,p)$
Q : solve(lhs,q)$
dz : rhs(P[1])*dx + rhs(Q[1])*dy;
sol: z = integrate(P[1],x) + integrate(Q[1],y)+%C$
print("The required solution is:")$
print(sol)$
```

Output :

```
eqn : p + x = q + y
dz : (k-x)dx + (k-y)dy;
The required solution is:
```

$$z = \frac{-y^2 + 2ky}{2} + \frac{-x^2 + 2kx}{2} + \%C$$

Program for type- IV:

```
/* Partial differential equation of type-IV */
kill(all) $
eqn: z = p*x + q*y + p*q ;
sol: subst([p=a,q=b], eqn) $
print(" The required soution is :",sol)$
```

Output :

```
z = px + qy + pq
The required soution is : by + ax + ab
```

Program for Lagrange's method :

```
/* PDE of type Pp + Qq = R using grouping method */
kill(all) $
P: x * y$
Q:y$
R:y * z$
eq: Pp + Qq = R;
print(" The A.E is") $
print(dx/P, "=", dy/Q, "=", dz/R) $
eq1:1/P = 1/Q * 'diff(y, x)$
eq2:1/P = 1/R * 'diff(z, x)$
print(" Both together the required solution ")$
ode2(eq1, y, x) $
subst(%c=a, %);
ode2(eq2, z, x)$
subst(%c=b, %);
```

Output :

xyp+yq=yz
The A.E is

$$\frac{dx}{xy} = \frac{dy}{y} = \frac{dz}{yz}$$

Both together the required solution

$y = \log(x) + a$
 $z = bx$

PROGRAM 5: Program to find the solution of total differential equation.

Program:

```
/* Program to solve a total differential equation */
kill(all)$
Z : (2*x*y + y^2)*dx + (x^2 + 2*x*y)*dy$
M : coeff(Z, dx)$
N : coeff(Z, dy)$
My : diff(M, y)$
Nx : diff(N, x)$
if My = Nx then (
    p : integrate(M, x),
    g : integrate(N - diff(p,y), y),
    sol : p + g,
    print("The given equation is exact"),
    print("The required solution is"),
    print(z = sol + %C)
)
else
    print("The given equation is not exact")$
```

Output - 1:

The given equation is exact
The required solution is:
 $z = xy^2 + x^2y + \%C$

Output - 2: ((3*x^2 + 2*y)*dx + (2*x + 4*y)*dy)

The given equation is exact
The required solution is:
 $z = 2y^2 + 2xy + x^3 + \%C$

PROGRAM 6: Program to get solutions of the partial differential equation using Charpit's method.

Program:

```
a)
/* Solution of PDE using Charpit's method */
kill(all)$
eq: x*p + y*q - p*q;
fx: diff(eq,x);
fy: diff(eq,y);
fz: diff(eq,z);
fp: diff(eq,p);
fq: diff(eq,q);
print("The Charpit's AE are")$
print(dp/(fx+ p*fz), "=", dq/(fy +q*fz), "=", dx/-fp , "=", dy/-fq , "=", dz/(-p*fp
- q*fq), "=dΦ/0")$
eq1:1/(fx +p*fz) * 'diff(p,q) = 1/(fy+ q*fz) $
ode2(eq1, p, q) $
k: subst(%c=a, %) $
subst(k,eq) $
qv: solve(%, q);
pv: subst(q=rhs(qv[1]), k);
s: z=rhs(pv)* dx + rhs(qv[1])*dy;
s: factor(s)$
s: subst(a*x+y=v, a*dx+dy=dv);
print ( "The required solution is")$
sol: a*z = integrate(rhs(coeff(s, dv)),v) + b ;
subst(v = (a*x+y)^2/2, sol);
```

Output:

$qy+px-pq$

p

q

0

$x-q$

$y-p$

The Charpit's AE are

$$dp/p = dq/q = dx/(q-x) = dy/(p-y) = dz/(-q*(y-p)-p*(x-q)) = d\Phi/0$$

$$[q=(y+ax)/a, q=0]$$

$$p=y+ax$$

$$z=(dy(y+ax))/a + dx(y+ax)$$

$$dy+adx=dv$$

The required solution is

$$az=v+b$$

$$az=((y+ax)^2)/2+b$$

b)

/* Solution of PDE using Charpit's method */

kill(all)\$

$$\text{eq: } p^2 + q^2 - z \text{ \$}$$

$$\text{fx: diff(eq,x);}$$

$$\text{fy: diff(eq,y);}$$

$$\text{fz: diff(eq,z);}$$

$$\text{fp: diff(eq,p);}$$

$$\text{fq: diff(eq,q);}$$

print("The Charpit's AE are")\$

$$\text{print(dp/(fx+ p*fz), "=", dq/(fy+q*fz), "=", dx/(-fp), "=", dy/(-fq), "=", dz/(-p*fp - q*fq), "=d\Phi/0")$}$$

$$\text{eq1:1/(fx+p*fz) * 'diff(p,q) = 1/(fy+q*fz) \$}$$

$$\text{ode2(eq1,p,q)$}$$

$$\text{k: subst(\%c=a, \%)\$}$$

$$\text{subst(k,eq) \$}$$

```

qv: solve(%, q);
pv: subst(q=rhs(qv[1]), k);
s: dz =rhs(pv)* dx + rhs(qv[1])*dy;
s: factor(s)$
print ( "The required solution is")$
sol: integrate(1/sqrt(z), z) = (integrate(-(a), x) + integrate(-(1), y))/ sqrt(a^2+1) + b
;

```

Output:

$$-z+q^2+p^2$$

$$0$$

$$0$$

$$-1$$

$$2p$$

$$2q$$

The Charpit's AE are

$$-dp/p = -dq/q = -dx/(2p) = -dy/(2q) = dz/(-2q^2-2p^2) = d\Phi/0$$

$$[q = -\sqrt{z}/\sqrt{a^2 + 1}, q = \sqrt{z}/\sqrt{a^2 + 1}]$$

$$p = -(a\sqrt{z}/\sqrt{a^2 + 1})$$

$$dz = -(dy\sqrt{z}/\sqrt{a^2 + 1}) - (adx\sqrt{z}/\sqrt{a^2 + 1})$$

The required solution is

$$2\sqrt{z} = (-y-ax)/\sqrt{a^2 + 1} + b$$

PROGRAM 7: Program to find the Laplace transforms of some standard and periodic functions.

Program:

```
a)
/* Laplace transform of some standard function */
kill(all)$
l:laplace(exp(a*t),t,s)$
print(" Laplace transform of exp(at) is", l)$
l1:laplace(cos(m*t),t,s)$
print(" Laplace transform of cos(mt) is", l1)$
l2: laplace(sin(m*t),t, s) $
print( "Laplace transform of sin(mt) is", l2)$
l3: laplace(cosh(m*t),t, s)$
print(" Laplace transform of cosh(mt) is", l3)$
l4: laplace(sinh(m*t),t,s)$
print("Laplace transform of sinh(mt) is", l4)$
l5: laplace(%e^(a* t) * cos(m*t),t, s)$
print("Laplace transform of e^(at) * cos(mt) is", l5)$
l6: laplace(%e^(a* t) * sin(m*t),t,s)$
print("Laplace transform of e^(at) * sin(mt) is", l6) $
l7: laplace(%e^(a* t) * cosh(m*t),t,s)$
print("Laplace transform of e^(at) *cosh(mt) is", l7)$
l8: laplace(%e^(a* t) * sinh(m*t),t,s)$
print("Laplace transform of e^(at) * sinh(mt) is", l8)$
assume (n>0)$
l9: laplace(t^n,t, s) $
print("Laplace transform of t^n is", l9) $
```

Output - 1:

Laplace transform of $\exp(at)$ is $\frac{1}{s-a}$

Laplace transform of $\cos(mt)$ is $\frac{s}{s^2 + m^2}$

Laplace transform of $\sin(mt)$ is $\frac{m}{s^2 + m^2}$

Laplace transform of $\cosh(mt)$ is $\frac{s}{s^2 - m^2}$

Laplace transform of $\sinh(mt)$ is $\frac{m}{s^2 - m^2}$

Laplace transform of $e^{at} * \cos(mt)$ is $\frac{s-a}{s^2 - 2as + m^2 + a^2}$

Laplace transform of $e^{at} * \sin(mt)$ is $\frac{m}{s^2 - 2as + m^2 + a^2}$

Laplace transform of $e^{at} * \cosh(mt)$ is $\frac{s-a}{s^2 - 2as - m^2 + a^2}$

Laplace transform of $e^{at} * \sinh(mt)$ is $\frac{m}{s^2 - 2as - m^2 + a^2}$

Laplace transform of t^n is $\frac{\gamma(n+1)}{s^{n+1}}$

b) Finding the Laplace transform of some periodic function

$$f(t) = 1 \text{ if } 0 < t < a/2$$

$$= -1 \text{ if } a/2 < t < a$$

/* Laplace transform of some periodic function */

```
kill(all)$
T:a;
f1(t) := 1;
f2(t):=-1;
k1: integrate(%e^-(s*t) * f1(t),t,0,T/2)$
k2: integrate(%e^-(s*t) * f2(t),t,T/2,T) $
K: 1/(1-%e^-(s*T)) * (k1+ k2) $
print("L[",f(t), "]= ",factor(radcan(K)))$
```

Output - 1:

a

f1(t) := 1

f2(t):= -1

$$L[f(t)] = \frac{e^{\frac{as}{2}} - 1}{s(e^{\frac{as}{2}} + 1)}$$

PROGRAM 8: Program to verify Convolution theorem.

Program:

```
a)
/* Verification of convolution theorem */
kill(all)$
assume(t>0) $
f(t):= t ;
g(t) := %e^t;
RHS: factor(laplace(f(t),t,s) * laplace(g(t),t,s));
K: integrate(f(u)*g(t-u),u,0,t);
LHS: factor(laplace(K,t, s));
if LHS = RHS then
    print(" Convolution theorem is verified")
else
    print(" Convolution theorem is not verified")$
```

Output - 1:

```
f(t) := t
g(t) := %e^t
1/(s-1)s^2
%e^t - t - 1
1/(s-1)s^2
Convolution theorem is verified
```

Output - 2: f(t) = g(t) := %e^t;

```
f(t) := %e^t
g(t) := %e^t
1/(s-1)^2
t%e^t
1/(s-1)^2
Convolution theorem is verified
```

PROGRAM 9: Program to solve ordinary linear differential equations using Laplace transforms.

Program:

a)

/* Solve ordinary linear differential equations using Laplace transforms */

kill(all)\$

eqn: 'diff(y,t,2) + 4* 'diff(y,t) + 4*y = %e^t ;

y1(k):=0\$

yd(k):=0\$

d1: s* L(y(t)) - y1(0)\$

d2: s^2*L(y(t)) - s*y1(0) - yd(0)\$

eq1: d2 + 4*d1 + 4*L(y(t)) = laplace(%e^t,t,s);

K: solve(eq1,L(y(t)));

print("The solution of ODE is y =", ilt(rhs(K[1]),s,t))\$

Output - 1:

$(d^2/dt^2)y + 4(d/dt)y + 4y = e^t$

$s^2L(y(t)) + 4sL(y(t)) + 4L(y(t)) = 1/(s-1)$

K [$L(y(t)) = 1/(s^3 + 3s^2 - 4)$]

The solution of ODE is $y = e^t/9 - t e^{-(2t)}/3 - e^{-(2t)}/9$

b)

/*Solve ordinary linear differential equations using Laplace transforms */

kill(all)\$

eqn: 'diff(y,t,2) + 3*'diff(y,t) + 2*y = sin(t)\$;

y1(k):=0\$

yd(k):=0\$

d1: s* L(y(t)) - y1(0)\$

d2: s^2*L(y(t)) - s*y1(0) - yd(0)\$

eq1: d2 + 3*d1 + 2*L(y(t)) = laplace(sin(t),t,s);

K: solve(eq1,L(y(t)));

print("The solution of ODE is y =", ilt(rhs(K[1]),s,t))\$

Output - 2:

$(d^2/dt^2)y + 3(d/dt)y + 2y = \sin(t)$

$s^2L(y(t)) + 3sL(y(t)) + 2L(y(t)) = 1/(s^2 + 1)$

K [L(y(t) = $1/(s^4+3s^3+3s^2+3s+2)$]

The solution of ODE is $y = \sin(t)/10 - 3\cos(t)/10 + \%e^{(-t)}/2 - \%e^{(-2t)}/5$

PROGRAM 10: Program to find full range Fourier series of some simple functions with period 2π and $2L$.

Program:

```
a)
/* Program to find full range Fourier series with period 2L */
kill(all)$
assume(L>0)$
f(x) := x$
a0 : (1/L)*integrate(f(x), x, -L, L);
an : (1/L)*integrate(f(x)*cos(n*%pi*x/L), x, -L, L);
bn : (1/L)*integrate(f(x)*sin(n*%pi*x/L), x, -L, L);
/*if not simplified then add below 2 lines
bn : subst([sin(%pi*n)=0, cos(%pi*n)=(-1)^n],bn)$
bn : ratsimp(bn); */
FS : a0/2 + sum(an*cos(n*%pi*x/L), n, 1, inf) + sum(bn*sin(n*%pi*x/L), n, 1,
inf)$
print("The full range Fourier series is:")$
print(FS)$
```

Output - 1:

```
a0 = 0
an = 0
bn = (2L(-1)^(n+1))/(n*%pi)
The full range Fourier series is:
x = (2L/%pi)Σ (-1)^(n+1)/n sin(n*%pi x/L)
```

b)

```
/* Full range Fourier series with period 2 */
kill(all)$
f(x) := x$
a0 : (1/%pi)*integrate(f(x), x, -%pi, %pi);
an : (1/%pi)*integrate(f(x)*cos(n*%pi*x/%pi), x, -%pi, %pi);
bn : (1/%pi)*integrate(f(x)*sin(n*%pi*x/%pi), x, -%pi, %pi);
/*if not simplified then add below 2 lines
bn : subst([sin(%pi*n)=0, cos(%pi*n)=(-1)^n],bn)$
bn : ratsimp(bn); */
FS : a0/2 + sum(an*cos(n*x), n, 1, inf) + sum(bn*sin(n*x), n, 1, inf)$
print("The Fourier series is:")$
print(FS)$
```

Output - 2:

$$a_0 = 0$$

$$a_n = 0$$

$$b_n = (2(-1)^{(n+1)})/(n)$$

The full range Fourier series is:

$$x = (2) \sum (-1)^{(n+1)}/n \sin(nx)$$

PROGRAM 11: Program to find half range sine and cosine series of some simple functions.

Program:

a)

```
/* Half Range Sine Series for f(x)=x^2 */
kill(all)$
declare(x, real)$
assume(0 < x, x < %pi)$
f(x) := x^2$
L : %pi$
bn : (2/L)*integrate(f(x)*sin(n*%pi*x/L), x, 0, L);
FS : sum(bn*sin(n*%pi*x/L), n, 1, inf)$
print("Half range sine series is:", FS)$
```

Output - 1:

$bn(n) = (2(-1)^n (2-n^2) \%pi)/n^3$
Half range sine series is:
 $\sum (2(-1)^n (2-n^2) \%pi)/n^3 \sin(nx)$

b)

```
/* Half Range Cosine Series for f(x)=x^2 */
kill(all)$
declare(x, real)$
assume(0 < x, x < %pi)$
f(x) := x^2$
L : %pi$
a0 : (2/L)*integrate(f(x), x, 0, L);
an : (2/L)*integrate(f(x)*cos(n*%pi*x/L), x, 0, L);
FS : a0/2 + sum(an*cos(n*%pi*x/L), n, 1, inf)$
print("Half range cosine series is:", FS)$
```

Output - 2:

$$a_0 = 2\pi^2/3$$
$$a_n = 4(-1)^n/n^2$$

Half range cosine series is:

$$\pi^2/3 + 4 \sum_{n=1}^{\infty} (-1)^n/n^2 \cos(nx)$$

PROGRAM 12: Program to find sine and cosine Fourier transforms.

Program:

a)

```
/* Fourier Sine Transform */
```

```
kill(all)$
```

```
assume(a > 0)$
```

```
f(x) := %e^(-a*x);
```

```
F_s : sqrt(2/%pi) * integrate(f(x)*sin(s*x), x, 0, inf)$
```

```
print("Fourier Sine Transform is:")$
```

```
print(F_s)$
```

Output - 1:

```
f(x) := %e^(-ax)
```

Fourier Sine Transform is:

$$\sqrt{\frac{2}{\pi}} \frac{s}{a^2 + s^2}$$

b)

```
/* Fourier Cosine Transform */
```

```
kill(all)$
```

```
assume(b > 0)$
```

```
f(x) := %e^(-b*x);
```

```
F_c : sqrt(2/%pi) * integrate(f(x)*cos(s*x), x, 0, inf)$
```

```
print("Fourier Cosine Transform is:")$
```

```
print(F_c)$
```

Output - 2:

```
f(x) := %e^(-bx)
```

Fourier Cosine Transform is:

$$\sqrt{\frac{2}{\pi}} \frac{b}{b^2 + s^2}$$